

Analytical MCQ Assessment

High-level cognitive evaluation paper focusing on mobile testing methodologies, tools, and challenges.

SECTION A: Multiple Choice Questions

Q1. When testing a mobile application, what is the primary technical difference between using an emulator and a simulator? (1 Marks)

- A. Emulators mimic both the hardware and software environments, while simulators mimic only the software behavior.
- B. Simulators mimic both the hardware and software environments, while emulators mimic only the software behavior.
- C. Emulators are used exclusively for iOS, while simulators are used exclusively for Android.
- D. Simulators run binary code directly, while emulators translate binary code at runtime.

Q2. During a mobile app performance test, you need to simulate a transition from a high-speed 5G network to a degraded 3G connection. Which testing technique is most appropriate to validate the app's resilience? (1 Marks)

- A. Boundary Value Analysis
- B. Network Throttling
- C. Static Code Analysis
- D. Load Balancing Simulation

Q3. A QA team is testing a hybrid mobile application. Which of the following issues is most unique to hybrid apps compared to native apps? (1 Marks)

- A. Memory leaks caused by native garbage collection.
- B. Inconsistent rendering of UI elements inside the WebView container.
- C. Inability to access device hardware like the camera or GPS.
- D. Failure to receive push notifications when the app is in the background.

Q4. In Appium automation, what is the primary purpose of using 'Desired Capabilities'? (1 Marks)

- A. To define the assertions and validation checkpoints for the test suite.
- B. To instruct the Appium server on the type of session, platform, and device configuration required.
- C. To generate HTML reports automatically after test execution.
- D. To manage the database connections during mobile integration testing.

Q5. An Android application crashes when the user rotates the screen from portrait to landscape while a data-fetching operation is active. What is the most likely cause of this defect? (1 Marks)

- A. The device ran out of physical storage space during rotation.
- B. The activity was destroyed and recreated without properly retaining the state or background

thread reference.

- C. The network connection was interrupted by the physical movement of the device.
- D. The application's manifest file lacked permission to access the accelerometer.

Q6. Which of the following practices is a critical security testing step to ensure that an Android APK cannot be easily reverse-engineered? (1 Marks)

- A. Verifying that the APK is signed with a debug certificate.
- B. Checking if the code has been obfuscated using tools like ProGuard or R8.
- C. Ensuring all network requests use HTTP instead of HTTPS.
- D. Confirming that the app's target SDK version is set to the lowest possible value.

Q7. A mobile app is flagged for high battery consumption during background idle states. Which testing tool or approach is best suited to diagnose this specific issue? (1 Marks)

- A. Functional regression testing using a cloud device farm.
- B. Analyzing wake locks and background CPU utilization using profiling tools like Android Profiler or Instruments.
- C. Running static analysis on the application's UI layout files.
- D. Executing unit tests on a local emulator.

Q8. When preparing an iOS application for external beta testing via TestFlight, which build configuration is mandatory? (1 Marks)

- A. The build must be signed with an Ad Hoc provisioning profile only.
- B. The build must be compiled using an Android-compatible wrapper.
- C. The build must be signed with an App Store Distribution provisioning profile.
- D. The build must have debugging enabled and be unsigned.

Q9. What is the primary objective of testing 'Deep Links' in a mobile application? (1 Marks)

- A. To verify that the app can establish a secure VPN connection.
- B. To ensure that clicking a specific URL redirects the user directly to a specific screen within the app.
- C. To measure the depth of the application's navigation hierarchy.
- D. To test the app's performance under deep database query loads.

Q10. How should a QA lead select the device matrix for compatibility testing of a new Android application to optimize coverage within a limited budget? (1 Marks)

- A. Test only on the latest flagship device from the market leader.
- B. Select a mix of devices based on target market share, OS versions, screen resolutions, and hardware specifications.
- C. Use emulators exclusively to avoid the cost of physical devices.
- D. Test on every Android device manufactured in the last ten years.

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: A

Rationale: Emulators replicate both hardware and software configurations, whereas simulators only mimic the software behavior of the target device.

Q2. Correct Answer: B

Rationale: Network throttling artificially limits bandwidth and increases latency to simulate real-world degraded network conditions.

Q3. Correct Answer: B

Rationale: Hybrid apps rely on WebViews to render web content, making them susceptible to rendering inconsistencies across different OS WebView engines.

Q4. Correct Answer: B

Rationale: Desired Capabilities are key-value pairs sent to the Appium server to configure the target device, platform, and application session.

Q5. Correct Answer: B

Rationale: Screen rotation triggers an activity destruction and recreation in Android, which can cause crashes if background tasks hold references to the destroyed activity.

Q6. Correct Answer: B

Rationale: Code obfuscation makes decompiled code difficult to understand, protecting the intellectual property and security of the APK.

Q7. Correct Answer: B

Rationale: Profiling tools allow testers to monitor CPU usage, network activity, and wake locks that keep the device awake and drain the battery.

Q8. Correct Answer: C

Rationale: TestFlight distribution requires the app to be signed with an App Store Distribution profile to ensure it meets Apple's security standards.

Q9. Correct Answer: B

Rationale: Deep link testing ensures that external URLs correctly route users to specific in-app content or screens.

Q10. Correct Answer: B

Rationale: A risk-based device matrix balancing market share, OS versions, and hardware profiles provides the most cost-effective compatibility coverage.